

Exam — Databases - Summer 2026

Exam Rules

Name: _____ Matriculation number: _____

1. Do not cheat.
2. Write your full name and matriculation number on the exam and each extra sheet.
3. You may use two double-sided A4 cheat sheets or four single-sided A4 cheat sheets.
4. Do not attach sticky notes to the cheat sheets.
5. If you need extra sheets, place them on your table before the exam starts.
6. Ask the exam supervisor before you use the restroom.
7. Use your floor's restroom only when it is vacant.
8. After the first student leaves the exam early, the remaining students cannot use the restroom.
9. Stay until the exam ends so that other students can use the restroom.
10. Hand in the exam and all extra sheets.

Task A – Explain Database Terms

10 P.

For each term, write 2-3 sentences. Include one concrete database example in these sentences.

- Primary key
- Foreign key
- Table
- Row
- SQL query

Task B – Multiple Choice: True or False

16 P.

For each statement, mark exactly one column. Each correct answer gives 2 points. Each wrong answer deducts 2 points. An empty row or two marks give 0 points for that row. The total score for this task cannot be negative.

Statement	True	False
A primary key identifies each row in a table.		
A foreign key can reference a primary key in another table.		
A table row is also called a record.		
The SELECT statement is used to delete rows from a table.		
The COUNT(*) function counts rows.		
A WHERE clause filters rows before they are returned.		
Every table must have a foreign key.		
NULL means the same as the number 0.		

Task C – Normalization and SQL Constraints

A small library uses one table and one row for each book in a loan. A loan can contain multiple books. Each *loan_id* identifies one member, loan date, and due date. Each *member_id* identifies one member name and email address. Each *book_id* identifies one title, author name, and genre. The *return_date* applies to one book in one loan.

```

1 CREATE TABLE library_loans (
2     loan_id INT,
3     loan_date DATE,
4     due_date DATE,
5     member_id INT,
6     member_name VARCHAR(100),
7     member_email VARCHAR(100),
8     book_id INT,
9     book_title VARCHAR(150),
10    author_name VARCHAR(100),
11    genre VARCHAR(50),
12    return_date DATE,
13    PRIMARY KEY (loan_id, book_id)
14 );

```

1. Identify three functional dependencies from the table description. Use this form: 6 P.

1 A -> B, C

A functionally determines B and C.

2. Describe two distinct anomalies or data redundancies. For each one, name the applicable schema columns. 6 P.
3. Decompose the table into 3NF relations. Mark primary-key attributes with (PK) and foreign-key attributes with (FK). Do not write SQL in this subtask. 14 P.
4. Write SQL CREATE TABLE statements for two relations from your 3NF design. Each statement must include a primary key and one other constraint. At least one statement must include a foreign key. 8 P.

Task D – Query Writing

Use the following relational schema:

```
1 Member(member_id, name, email)
2 Book(book_id, title, genre)
3 Loan(loan_id, member_id, book_id, loan_date, return_date)
```

SQL

The first attribute in each relation is its primary key. *Loan.member_id* references *Member.member_id*.
Loan.book_id references *Book.book_id*.

1. Return the name and email of each member who has an open loan. Return each member once. A loan is open when *return_date* IS NULL. 6 P.
2. Return the total number of loans for each book genre that has at least one loan. Return genre and *loan_count*. 8 P.
3. Return *member_id*, *name*, and *email* for each member who has never borrowed a book. 8 P.

Task E — Database Design: Course Registration System

Design a database for the following restricted course registration scenario.

A university has departments. Each department has a unique name and offers courses. Each course has a globally unique course code and a title. Each course belongs to exactly one department. Each lecturer has a name and works for exactly one department.

A course can have multiple offerings, such as offerings in different semesters. Each course offering has one course, one semester, one year, and exactly one lecturer.

Students can enroll in course offerings. Each student has a name and a unique matriculation number. A student email address is optional and unique when present. An enrollment has an enrollment date, a status, and an optional grade. The valid status values are *active*, *completed*, and *withdrawn*.

Use surrogate integer IDs where useful. You may add attributes, but your design must contain all specified information.

1. Draw or describe the ER model. Include the required entity types, relationships, and cardinalities. Write cardinalities explicitly if your drawing is ambiguous. 12 P.

Grading checklist: entities (4 P), required attributes and identifiers (3 P), cardinalities (3 P), handling repeated course offerings and enrollments (2 P).

2. Transform your ER model into a relational model. Write each relation as Table(attribute, attribute, ...). Mark primary keys with (PK) and foreign keys with (FK). Mark unique attributes with (UNIQUE) and optional attributes with (OPTIONAL). 14 P.

Grading checklist: correct tables (4 P), primary keys (3 P), foreign keys (4 P), optional values and uniqueness constraints (2 P), no unnecessary duplicated department/lecturer/student data (1 P).

3. Write PostgreSQL CREATE TABLE statements for CourseOffering and Enrollment. Include primary keys, foreign keys, and constraints for all required values. 10 P.

Grading checklist: syntactically valid DDL (2 P), PKs (2 P), FKs (3 P), **NOT NULL**/optional values handled correctly (2 P), useful status or year constraint (1 P).

4. Return one row for each active enrollment. Include the student name, course title, and lecturer name. An enrollment is active when status = 'active'. 8 P.

Grading checklist: correct joins (4 P), filter for active enrollments (2 P), selected output columns and aliases are understandable (2 P).

In total, you can achieve 126 points. You have achieved ____ P. of 126 points.

Points	126–113	112–100	99–88	87–75	74–64	63–0
Value	A	B	C	D	E	F